



Mathematical Model

Part 3

Integrated Provisioning Services
Airbus Services Solutions

TABLE OF CONTENTS

1.	INTRODUCTION	2
2.	MATHEMATICAL PRINCIPLES.....	3
2.1.	BASIC FORMULAS	3
2.2.	CUSTOMER PARAMETERS	3
2.3.	PART and AIRCRAFT PARAMETERS	3
2.4.	FURTHER VARIABLES USED	4
2.5.	POISSON DISTRIBUTION	4
3.	APPLICATION OF THE POISSON DISTRIBUTION.....	5
3.1.	MANUAL DETERMINATION OF RECOMMENDED QUANTITY.....	5
3.2.	HOW TO USE THE POISSON DISTRIBUTION TABLE.....	6
3.3.	POISSON TABLES.....	7

1. INTRODUCTION

This brochure explains to you the process of the standard Mathematical Model that is used to determine recommended spare parts quantities.

Once you go through the examples provided hereafter, you will see that the model is just straightforward.

Since this model is nothing but basic, we have developed other methods of provisioning recommendations which go beyond the range of these ground rules. However, understanding the groundwork helps in any case to identify the major cost driving factors of the initial spares investment and the link of these factors to the operational profile of the airlines.

These basic understandings will also enable you to pose the correct questions leading to reduction of your initial provisioning investment.



Tamas Stefka
Director Integrated Provisioning Services

2. MATHEMATICAL PRINCIPLES

2.1. BASIC FORMULAS

Generically a part is analysed for its risk of failure within the first year of operation via testing the annual demand (D_{ann}) versus the minimum annual demand (MAD). In case $D_{ann} \geq MAD$ the first spare is allocated to the stock. Further risks out of re-supply chain delays are mitigated by adding further spares to obtain sufficient protection in line with the expected demand (D_{rst}) during the re-supply time.

Annual Demand:

$$D_{ann} = \frac{FH \times FS \times QPA}{MTBUR}$$

Expected Demand during Re-Supply Time:

$$RST = \left[\left((MST + TT) \times \left(1 - \frac{SCR}{1000} \right) \right) + \left((LTM + AT) \times \left(\frac{SCR}{1000} \right) \right) \right]$$

$$D_{RST} = \frac{D_{ann}}{365} \times RST$$

2.2. CUSTOMER PARAMETERS

AT	Administration Time
FC	average Flight Cycle time
FH	Flight Hours per aircraft per year
MAD	Minimum Annual Demand
FS	Number of A/C in the IP period
TT	Transit Time

2.3. PART and AIRCRAFT PARAMETERS

QPA	Quantity per aircraft
SCR	Scrap Rate
LTM	Lead Time
MST	Max/Mean shop processing time
TAT	Turn-Around-Time (= TT + MST)
MCBUR	Mean Cycles Between Unscheduled Removals
MTBUR	Mean Time Between Unscheduled Removals
	For certain items the operation is measured in cycles. In these cases the MTBUR is calculated as follows: $MTBUR = MCBUR \times FC$.

2.4. FURTHER VARIABLES USED

D_{ann}	Annual Demand
D_{rst}	Expected Demand during Re-Supply Time
m	Recommended Quantity
PL	Spares Protection Level may vary from 80.0% to 98.0% and may be selected as function of equipment's Essentiality Code (ESS) and Spare Part Class (SPC) or as function of ESS and 3 different price ranges (Low, Medium, High) where range limits are defined by the customer
PLT	Tolerance used when testing PL obtained from m versus PL requested by customer may vary from 0 to 10 percent points
CRF	Customised Redundancy Factor may vary from 0.0 to 1.0 (0% to 100%) with Go If equipment when $n > 1$. Captures the amount of system redundancy a customer intends to consider in spares planning. CRF = 0 for No Go and Go Equipment

2.5. POISSON DISTRIBUTION

Recommended quantities (m) and Protection Levels (PL) are now assessed assuming a Poisson distribution of part failures across time. In a Poisson process m cannot be calculated straightforwardly. It requires an iterative process of assuming m , testing it against the PL requirement and incrementing m to test it again, etc.

Gauss distribution is applied for an annual demand ≥ 127 .

Formula

$$P\{R \leq m\} = e^{-D_{rst}} \times \sum_0^m \frac{(D_{rst})^m}{m!}$$

P = Probability
R = number of removals

The D_{ann} versus MAD test is assumed to have the part qualified as spare. Then the formula describes the probability to have no more removals than spares during the re-supply time

Whereas above may be done by programmable IT tools, the following page illustrates how m may be obtained manually from tables showing P against various m to a given E_r .

Remark:

When the customer selects the Price Range option, parts lacking a price are considered to have a medium range price.

For Parts, where LTM data is not available, LTM defaults to present values depending on the type of material.

<u>Type of material</u>	<u>LTM [days]</u>
Airbus Proprietary Parts	10
Supplier Equipment	60
Supplier Equipment Breakdown Parts	60
Standard Hardware	30

Others

90

3. APPLICATION OF THE POISSON DISTRIBUTION

3.1. MANUAL DETERMINATION OF RECOMMENDED QUANTITY

See as well the table(s) on next the page(s)

Case A: MAD = 1,0 $D_{ann} = 0,50$

- D_{ann} is less than MAD → part does not qualify as spare → **m = 0**
! Probability of failure is considered too low

Case B: MAD = 1,0 $D_{ann} = 2,0$ PL = 98% $D_{rst} = 1,1$

- $D_{ann} \geq MAD$ → part qualified as spare → **m = 1**
- PL (m=1) = 0.699 = 69,9% is below PL → increment by 1
- PL (m=2) = 0.900 = 90,0% is below PL → increment by 1
- PL (m=3) = 0.974 = 97,4% is below PL → increment by 1
- PL (m=4) = 0.996 = 99,6% is above PL → **m=4**

Case C: as Case B above but PLT = 1%

- PL (m=3) = 0.974 = 97,4% is above PL-PLT = 97% → **m=3**
! The tolerance reduces stocks by 1 spare

Case D: as Case B above but PLT = 1% and CRF = 100%

(Part is Go If and n=2)

- The D_{rst} calculation needs to consider the factor $\frac{1}{1+CRF}$
- New D_{rst} is then $\frac{D_{rst}}{1+1} = 0,55$ (see table on page 7)
- PL (m=1) = 0.894 = 89,4% is below PL-PLT → increment by 1
- PL (m=2) = 0.982 = 98,2% is above PL-PLT → **m=2**
! The redundancy reduces stocks by 2 units

3.2. HOW TO USE THE POISSON DISTRIBUTION TABLE

Sample table for $D_{rst} = 1.0$ to 2.0

ESTIMATED
DEMAND [D_{rst}]

m	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	.368	.383	.301	.273	.247	.223	.202	.183	.165	.150	.135
1	.736	.699	.663	.627	.592	.558	.525	.493	.463	.434	.406
2	.920	.900	.879	.857	.833	.809	.783	.757	.731	.704	.677
3	.981	.974	.966	.957	.946	.934	.921	.907	.891	.875	.857
4	.996	.996	.992	.989	.986	.981	.976	.970	.964	.956	.947
5	.999	.999	.998	.998	.997	.996	.994	.992	.990	.987	.983
6	1.000	1.000	1.000	1.000	.999	.999	.999	.998	.997	.997	.995
7					1.000	1.000	1.000	1.000	.999	.999	.999
8									1.000	1.000	1.000

RECOMMENDED
QUANTITY [m]

PROTECTION
LEVEL [PL]

3.3. POISSON TABLES

	Estimated Demand										
m	.02	.04	.06	.08	.10	.15	.20	.25	.30	.35	.40
0	.980	.961	.942	.923	.905	.861	.819	.779	.741	.705	.670
1	1.000	.999	.998	.997	.995	.990	.982	.974	.963	.951	.938
2		1.000	1.000	1.000	1.000	.999	.999	.998	.996	.994	.992
3						1.000	1.000	1.000	1.000	1.000	.999
4											1.000

Table 1

	Estimated Demand										
m	.45	.50	.55	.60	.65	.70	.75	.80	.85	.90	.95
0	.638	.607	.577	.549	.522	.497	.472	.449	.427	.407	.387
1	.925	.910	.894	.878	.861	.844	.827	.809	.791	.772	.754
2	.989	.986	.982	.977	.972	.966	.959	.953	.945	.937	.929
3	.999	.998	.998	.997	.996	.994	.993	.991	.989	.987	.984
4	1.000	1.000	1.000	1.000	.999	.999	.999	.999	.998	.998	.997
5					1.000	1.000	1.000	1.000	1.000	1.000	1.000

POISSON TABLES (continued)

	Estimated Demand										
m	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	.368	.333	.301	.273	.247	.223	.202	.183	.165	.150	.135
1	.736	.699	.663	.627	.592	.558	.525	.493	.463	.434	.406
2	.920	.900	.879	.857	.833	.809	.783	.757	.731	.704	.677
3	.981	.974	.966	.957	.946	.934	.921	.907	.891	.875	.857
4	.996	.996	.992	.989	.986	.981	.976	.970	.964	.956	.947
5	.999	.999	.998	.998	.997	.996	.994	.992	.990	.987	.983
6	1.000	1.000	1.000	1.000	.999	.999	.999	.998	.997	.997	.995
7					1.000	1.000	1.000	1.000	.999	.999	.999
8									1.000	1.000	1.000

POISSON TABLES (continued)

	Estimated Demand										
m	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8	4.0	4.2
0	.111	.091	.074	.061	.050	.041	.033	.027	.022	.018	.015
1	.355	.308	.267	.231	.199	.171	.147	.126	.107	.092	.078
2	.623	.570	.518	.469	.423	.380	.340	.303	.269	.238	.210
3	.819	.779	.736	.692	.647	.603	.558	.515	.473	.433	.395
4	.928	.904	.877	.848	.815	.781	.744	.706	.668	.629	.590
5	.975	.964	.951	.935	.916	.895	.871	.844	.816	.785	.753
6	.993	.988	.983	.976	.966	.955	.942	.927	.909	.889	.867
7	.998	.997	.995	.992	.988	.983	.977	.969	.960	.949	.936
8	1.000	.999	.999	.998	.996	.994	.992	.988	.984	.979	.972
9		1.000	1.000	.999	.999	.998	.997	.996	.994	.992	.989
10				1.000	1.000	1.000	.999	.999	.998	.997	.996
11							1.000	1.000	.999	.999	.999
12									1.000	1.000	1.000

POISSON TABLES (continued)

	Estimated Demand										
m	4.4	4.6	4.8	5.0	5.2	5.4	5.6	5.8	6.0	6.2	6.4
0	.012	.010	.008	.007	.006	.005	.004	.003	.002	.002	.002
1	.066	.056	.047	.040	.034	.029	.024	.021	.017	.014	.012
2	.185	.163	.143	.125	.109	.095	.082	.072	.062	.054	.049
3	.359	.326	.294	.265	.238	.213	.191	.170	.151	.134	.119
4	.551	.513	.476	.440	.406	.373	.342	.313	.285	.259	.235
5	.720	.686	.651	.616	.581	.546	.512	.478	.446	.414	.384
6	.844	.818	.791	.762	.732	.702	.670	.638	.606	.574	.542
7	.921	.905	.887	.867	.845	.822	.797	.771	.744	.716	.687
8	.964	.955	.944	.932	.918	.903	.886	.867	.847	.826	.803
9	.985	.980	.975	.968	.960	.951	.941	.929	.916	.902	.886
10	.994	.992	.990	.986	.982	.977	.972	.965	.957	.949	.939
11	.998	.997	.996	.995	.993	.990	.988	.984	.980	.975	.969
12	.999	.999	.999	.998	.997	.996	.995	.993	.991	.989	.986
13	1.000	1.000	1.000	.999	.999	.999	.998	.997	.996	.995	.994
14				1.000	1.000	1.000	.999	.999	.999	.998	.997
15							1.000	1.000	.999	.999	.999
16									1.000	1.000	1.000

POISSON TABLES (continued)

m	Estimated Demand											
	6.6	6.8	7.0	7.2	7.4	7.6	7.8	8.0	8.5	9.0	9.5	10.0
1	.010	.009	.007	.006	.005	.004	.004	.003	.002	.001	.001	.000
2	.040	.034	.030	.025	.022	.019	.016	.014	.009	.006	.004	.003
3	.105	.093	.082	.072	.063	.055	.048	.042	.030	.021	.015	.010
4	.213	.192	.173	.156	.140	.125	.112	.100	.074	.055	.040	.029
5	.355	.327	.301	.276	.253	.231	.210	.191	.150	.116	.089	.067
6	.511	.480	.450	.420	.392	.365	.338	.313	.256	.207	.165	.130
7	.658	.628	.599	.569	.539	.510	.481	.453	.386	.324	.269	.220
8	.780	.755	.729	.703	.676	.648	.620	.593	.523	.456	.392	.333
9	.869	.850	.830	.810	.788	.765	.741	.717	.653	.587	.522	.458
10	.927	.915	.901	.887	.871	.854	.835	.816	.763	.706	.645	.583
11	.963	.955	.947	.937	.926	.915	.902	.888	.849	.803	.752	.697
12	.982	.978	.973	.967	.961	.954	.945	.936	.909	.876	.836	.792
13	.992	.990	.987	.984	.980	.976	.971	.966	.949	.926	.898	.864
14	.997	.996	.994	.993	.991	.989	.986	.983	.973	.959	.940	.917
15	.999	.998	.998	.997	.996	.995	.993	.992	.986	.978	.967	.951
16	.999	.999	.999	.999	.998	.998	.997	.996	.993	.989	.982	.973
17	1.000	1.000	1.000	.999	.999	.999	.999	.998	.997	.995	.991	.986
18				1.000	1.000	1.000	1.000	.999	.999	.998	.996	.993
19								1.000	.999	.999	.998	.997
20									1.000	1.000	.999	.998
21											1.000	.999
22												1.000

POISSON TABLES (continued)

m	Estimated Demand											
	11	12	13	14	15	16	17	18	19	20	21	22
11	.579	.462	.353	.260	.185	.127	.085	.055	.035	.021	.013	.008
12	.689	.576	.463	.358	.268	.193	.135	.092	.061	.039	.025	.015
13	.781	.682	.573	.464	.363	.275	.201	.143	.098	.066	.043	.028
14	.854	.772	.675	.570	.466	.368	.281	.208	.150	.105	.072	.048
15	.907	.844	.764	.669	.568	.467	.371	.287	.215	.157	.111	.077
16	.944	.899	.835	.756	.664	.566	.468	.375	.292	.221	.163	.117
17	.968	.937	.890	.827	.749	.659	.564	.469	.378	.297	.227	.169
18	.982	.963	.930	.883	.819	.742	.655	.562	.469	.381	.302	.232
19	.991	.979	.957	.923	.875	.812	.736	.651	.561	.470	.384	.306
20	.995	.988	.975	.952	.917	.868	.805	.731	.647	.559	.471	.387
21	.998	.994	.986	.971	.947	.911	.861	.799	.725	.644	.558	.472
22	.999	.997	.992	.983	.967	.942	.905	.855	.793	.721	.640	.556
23	1.000	.999	.996	.991	.981	.963	.937	.899	.849	.787	.716	.637
24		.999	.998	.995	.989	.978	.959	.932	.893	.843	.782	.712
25		1.000	.999	.997	.994	.987	.975	.955	.927	.888	.838	.777
26			1.000	.999	.997	.993	.985	.972	.951	.922	.883	.832
27				.999	.998	.996	.991	.983	.969	.948	.917	.877
28				1.000	.999	.998	.995	.990	.980	.966	.944	.913
29					1.000	.999	.997	.994	.988	.978	.963	.940
30						.999	.999	.997	.993	.987	.976	.959
31						1.000	.999	.998	.996	.992	.985	.973
32							1.000	.999	.998	.995	.991	.983
33								1.000	.999	.997	.994	.989

© AIRBUS S.A.S. All rights reserved. Confidential and proprietary document.

This document and all information contained herein is the sole property of AIRBUS S.A.S. No intellectual property rights are granted by the delivery of this document or the disclosure of its content. This document shall not be reproduced or disclosed to a third party without the express written consent of AIRBUS S.A.S.

This document and its content shall not be used for any purpose other than that for which it is supplied. The statements made herein do not constitute an offer. They are based on the mentioned assumptions and are expressed in good faith. Where the supporting grounds for these statements are not shown, AIRBUS S.A.S. will be pleased to explain the basis thereof.

AIRBUS, its logo, A300, A310, A318, A319, A320, A321, A330, A340, A350, A380, A400M are registered trademarks



Airbus Services Solutions
Integrated Provisioning Services
Weg beim Jäger 150
22335 Hamburg, Germany
email: material-readiness.provisioning@airbus.com
Phone: +49.40.50 76 0